

Programme	B. Sc. Computer Science Minor				
Course Title	Emerging Trends in Computer Science				
Type of Course	Minor				
Semester	III				
Academic Level	200-299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	1. Knowledge in Computers. 2. Basic knowledge in Internet				
Course Summary	This course provides an overview of the latest trends and advancements in the field of computer science. Students will explore emerging technologies, methodologies, and research areas shaping the future of computing.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
		U	C	
CO1	Analyze real-world use cases and applications of emerging technologies, identifying opportunities and challenges for innovation and problem-solving in areas such as healthcare, finance, smart cities, and industry.	An	C	
CO2	Understand the fundamental concepts of artificial intelligence (AI), and applications across various domains.	U	C	
CO3	Identify the key components of a block chain network, such as nodes, blocks, transactions, and smart contracts.	Ap	C	

CO4	Understand the fundamental concepts of computer networks and popular applications	U	C	
CO5	Identify the key components of a block chain network and applications	Ap	C	
CO6	Understand the evolution of database management systems (DBMS) from traditional modern .	U	C	
CO7	Describe the features of NoSQL databases and their advantages over traditional relational databases.	Ap	C	
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

References:

Module	Unit	Content	Hrs	Marks(70)
I	Basic concepts of Artificial Intelligence and Machine Learning		12	20
	1	Concept of Machine Learning and Artificial Intelligence: Definition, Evolution	2	
	2	Types Of Machine Learning: Supervised learning, Unsupervised learning, Reinforcement learning, Evolutionary learning	3	
	3	Common ML algorithms: Regression, Classification, Clustering. (Concepts)	3	
	4	The Machine Learning Process: Data Collection and Preparation, Feature Selection , Algorithm Choice , Parameter and Model Selection , Training , Evaluation	2	
	5	Application of Machine Learning: Healthcare, Finance, Self Driving Cars, Robotics	2	
II	Introduction to Block chain Technology:		12	20

	6	Cryptography Overview: Definition, Types of Cryptography -Public and Private Keys, Application -Digital Signature	3	
	7	Introduction to Block chain Technology: History of Block chain, Generic Elements of Block chain, Features of Block chain(Decentralization, Transparency , Immutability and Security)	2	
	8	Types of Block chain:	2	
	9	Applications of Bock chain Technology : Financial Services, Supply Chain Management, Smart Contracts	2	
	10	Crypto currencies: Definition, Bit coin, Ethereum	2	
	11	Challenges in Block chain Adoption: Scalability, interoperability, and regulatory concerns, Security considerations.	1	
III	IOT and Cloud Technology		10	15
	12	Overview of Computer Networks: Definition, types, and importance.	2	
	13	Network Design Concepts: ISO/OSI and TCP/IP	2	
	14	Networking Devices and Protocols: Routers, switches, Hub, Modems, TCP, UDP, IP.	2	
	15	Cloud Computing and Services: Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS).	2	
	16	Internet of Things: Definition, Key characteristic, Architecture and components, Challenges and Security in IOT.	2	
IV	Unstructured Database		10	15
	17	Overview of traditional DBMS: Relational, Object-Oriented Database	3	
	18	Structured and Unstructured Database	1	
	19	Introduction to NoSQL databases	2	
	20	Types of NoSQL databases (Document-oriented, Key-value stores, Column-family stores, Graph databases)		
	21	Cloud-Based Database Services: Database as a Service (DBaaS) overview, Benefits of cloud-based solutions	2	

	22	Block chain Databases (2 hours): Understanding graph databases and their applications, Overview of Block chain databases and their role in data integrity.	2	
V	Practical Applications, Case Study		30	
	1	1. Identify the various software platforms for AI programming. 2. Identify the various platforms used for No-Code AI.(Google Cloud Auto ML, Microsoft Azure AI Buildeer, IBM Watson Studio etc..) 3. Use chatbot platforms like ChatGPT or any other to engage in conversational interactions and understand how natural language processing works. 4.Use online tools or any applications that demonstrate image recognition capabilities. 5.Use online platforms or software that provide interactive AI demos and simulations, such as neural network visualisers or AI-powered character generators. 6.Use an online tool like CyberChef or an online RSA key generator to generate a pair of RSA public and private keys. 7. Create a digital signature for a given document using an online service like DocuSign or HelloSign, and verify it using the service's verification features. 8. Use online resources to create a comparison table for public, private, and consortium blockchains, including real-world examples. 9. Use MySQL or PostgreSQL to create a database, define tables, and perform CRUD operations. 10.Use db4o (Database for Objects) or an equivalent tool to create and manipulate an object-oriented database. 11. Use MongoDB to store and query unstructured data. 12. Use MongoDB to create a collection and perform CRUD operations. 13. Deploy and interact with a cloud-based database service.		

		14. Explore and implement a basic blockchain database using an appropriate platform.		
		15. Use BigchainDB or a similar blockchain database platform to create a blockchain database.		

1. “Explorations in Artificial Intelligence and Machine Learning” - By Roberto Zicari
2. “Blockchain Fundamentals”- Dr. Ravindhar Vadapalli
3. "Cloud Computing: Concepts, Technology & Architecture" by Thomas Erl, Ricardo Puttini, and Zaigham Mahmood
4. Michael Miller, “The Internet of Things: How Smart TVs, Smart Cars, Smart Homes, and Smart Cities Are Changing the World”, Pearson Education 2015
5. “A Brief Guide to the Emerging World of Polyglot Persistence:- By Pramod J. Sadalage, Pramod Sadalage, Martin Fowler

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	-	-	1	-	-	3						
CO 2	-	2	1	-	2	3						
CO 3	-	2	1	-	2	3						
CO 4	-	2	1	-	2	3						
CO 5	-	2	1	-	2	3						
CO 6	-	2	1	-	2	3						
CO7	-	-	1	-	-	3						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate /

	Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2	✓			✓
CO 3	✓	✓		✓
CO 4		✓		✓
CO 5		✓		✓